Defect workflow

Read each column from top to bottom, A to G: F and G share the last one. Diamonds are decisions, and every branch out of one is labelled; "given" means set in the project file. Dashed steps are optional and off the main path. One command runs every step in order: `python3 dwf.py <name>.dwf`. Each phase writes summary.txt (for you to read) and results.json (for the later phases) in its own folder.

start / end · done by dwf.py itself · CASTEP jobs (run through supercell_convergence.py) · MACE (B2, C1: a foundation model; D3–D5, G2: one trained per defect) · ShakeNBreak · needs your attention · decision · input or data · optional, off the main path · (B) continues in the next column

A Host settings and reference data

A_host/ · once per host

project file <name>.dwf, host .cell and .param, jobscript

Converge k-points and cut-off? (converge) — no → Use your .param and k-points as they are

yes

A1 Converge k-points and cut-off
host energy per atom on each k-point grid (at a low cut-off), then for each cut-off in converge_cutoffs

Has the energy settled, within converge_tol? — no → Not settled: add finer grids or higher cut-offs, then rerun

yes

A2.1 Relax the host
lattice and positions, at A1's settings; every later cell is built from it

A2.2 Band edges
a single point on a dense Γ-centred grid: VBM, CBM and band gap

Dielectric constant given? (dielectric) — yes → Use that value

no

A2.3 Dielectric constant
DFPT at Γ (phonon+efield): ε_0 and ε_∞; ε_0 screens every charge correction

Chemical potentials given? — yes → Use them (%block chemical_potentials)

no

Competing phases given? (phases, or phases_from) — no → No limits: formation energies without μ terms

yes

A3.1 Relax the competing phases
yours, or the Materials Project's (doped), at the same settings as the host

Is the host stable against them? — no → Stop: check those phases and their settings

yes

A3.2 Chemical-potential limits
μ at each corner of the host's stability region (pymatgen); each dopant's μ is the highest its own phases allow

(B)

B Candidates, the screening and electronic cells

B_candidates/ · once per host

(A)

Defects listed in %block defects? — yes → **B1 Your defects**
one candidate per line; a pair with all: every arrangement of it

no

B1 List the candidates
vacancies and dopant substitutions on each inequivalent site (antisites and interstitials too, if switched on), then dopant–native pairs in every distinct arrangement up to pair_max_distance

B2 Supercell sizes
for each length in scan_lengths, the most compact supercell whose periodic images are at least that far apart (found by doped, or n × your transform)

Test each candidate's size with MACE? (mace and scan) — no → No test: B3 goes by the Γ-only floor alone

yes

B2 MACE size test
each candidate relaxed in every size it fits; it needs the smallest size whose relaxation energy matches every bigger size's within scan_tol, with its edge atoms moving less than scan_edge_tol

Screening cell given? (supercell) — yes → Use it (B2's test is then only reported)

no

B3 Choose the screening cell
the smallest size that is big enough for every candidate (the worst case decides) and at least the Γ-only floor (note below)

Do the host's band edges (A2.2) fold onto Γ of this cell? — no → Warning: Γ-point screening sees a wider gap (C2 measures it); B4's electronic cell folds the edges

yes

B4 The electronic cell (C4 on)
the smallest size, from the screening cell up, that folds every band-edge valley onto Γ (or electronic_cell)

(C)

The Γ-only floor (B3, B4)
Γ-point runs see a defect's levels as an isolated defect has them only when its periodic images barely overlap. A level E_b inside the gap decays as $\exp(-r/\xi)$, where (two-band model) $1/\xi^2 = (2m^*/\hbar^2)\, E_b\,(1 - E_b/E_g)$. A cell's images are at least decay_length_multiple × ξ apart (4ξ by default), with
- E_b = deep_level_margin (B3) or electronic_depth (B4)
- m^*: the lighter of the hole and electron masses at A2.2's band edges
- E_g: A2.2's band gap

Set screen_min_image to give B3's floor yourself (0: no floor).

Cell sizes are image distances
The shortest distance between a defect and its periodic images (Å), from scan_lengths: B2 tests every candidate in every size, B3's screening cell fits the worst case and the Γ-only floor, and B4's electronic cell (C4 on) also folds the band edges onto Γ.

C Screening

C_screen/ · every candidate, then the survivors in the electronic cell

(B)

C1 Build and pre-relax
every candidate in the screening cell, pre-relaxed with MACE (reusing B2's)

Filter by MACE energy? (configuration_margin) — yes → Drop arrangements more than that above the lowest MACE energy of their composition

no

C2 Screen in CASTEP
a cheap Γ-point relaxation of every candidate in each of screen_charges, from a triplet (a doublet for an odd electron count), plus the pristine cell

C3: passes the hull (charge-corrected), duplicate and property filters? — no → Dropped (the reason is in summary.txt)

yes

Into the electronic cell (B4)
everything after C3 works in it: each survivor's screened structures carried over (distort_cell) and relaxed in C5

C4 Distort (ShakeNBreak)
in the electronic cell, each charge state C3 carries on: neighbours moved in and out by each of distortions, others rattled

C5 Relax the distortions
and each start, at Γ with C2's settings, against the electronic cell's band edges; every step kept: D1's training data

(D)

What C keeps for D1–D5
Every geometry step of C5, with its energy and forces, in each .castep: D1 turns them into each defect's training data (the pbe head, at the screening settings), and D2 recomputes a tenth of them with the hybrid, in the electronic cell like everything after C3. With model : false, D6 ranks C5's relaxations instead.

The model in D1–D5: Wang et al.
Wang, Mosquera-Lois and Walsh, PRX Energy 2026 (arXiv:2603.05238): Sb_2Se_3, one 60-atom cell, VASP (PAW, 350 eV, DFT-D3).
- $V_{Se}(1)$ in q = −2 … +2: one charge-embedded MACE, trained on HSE06 Γ-point relaxations of ShakeNBreak distortions (2,287 structures)
- neutral $V_{Se}(1)$: a multi-fidelity MACE from 452 PBE steps and 34 HSE06 single points
- MACE 0.3.15 from scratch: 2 layers, 128 channels, 5.5 Å cut-off, ZBL
- test errors 0.5 meV/atom and 20 meV/Å; transition levels within 0.015 eV of HSE06; the search about 1,000 × cheaper than HSE06
- one model per defect, site and cell size; negative charge states were the hardest; the charge embedding and the two heads (D3) were never tested together: they trained them as two models

Data and trained models: Zenodo, record 22756145 (CC BY 4.0).

Foundation models on defects (their test)
On $V_{Sb}(1)$ at the PBE level, MACE-MH-1, MACE-MPA-0, GRACE-2L-OMAT and MatterSim put the minima 0.2–0.4 Å from DFT's (0.1 Å counts as a match), on flatter energy surfaces, and DFT relaxations from their minima missed the DFT ground states. The charge-aware MACE-POLAR-1-L and QET-PES-MatQ failed too. So B2 and C1 keep a foundation MACE for cell sizes and pre-relaxation only, and each defect gets a model trained on its own DFT data (D3).

What the filters in C and D keep
- C1 configuration_margin: arrangements within that of the lowest MACE energy of their composition (MACE ignores the charge)
- C3 hull: an arrangement stays if, at some Fermi level in the gap, its lowest formation energy is within hull_margin of its group's
- C3 duplicates: one that relaxed into a lower one's structure goes
- C3 properties (optional), met by a charge state stable somewhere in the gap: require_gap_states, require_spin, require_bound
- D6: the keep_structures lowest structures of each charge state

D Ground and excited states

D_states/ · the survivors, in the electronic cell

(C)

Search with a model of each defect? (model) — no → (skip)

yes

D1 Training data
each defect's C5 geometry steps and rattled pristine cells, in the electronic cell; filtered, a fifth kept for testing

D2 Hybrid single points
HSE06 (norm-conserving) on about a tenth of them, in the electronic cell: first, last and an even sample

D3 Train a model per defect
MACE from scratch: pbe and hse heads on one representation, the total charge embedded (every charge state)

D4: accurate? (test errors; the hybrid's forces at its minima) — no → Its minima's single points join the data; train again (at most model_rounds)

yes

D5 Search, electronic cell
ShakeNBreak distortions (±0.7) and rattles in every charge state, relaxed with the hse head; each minimum tried in the other charge states

D6 Rank the ground states
the model's lowest minima, or (no model) C5's, sharing distortions; keep_structures go on

D8 Validate
the screening cell vs bigger cells; Γ vs k-points (for you)

D7 Excited states (ΔSCF)
electronic cell, Γ: ground states relaxed; triplets only; localised levels (participation numbers); ZPL, E_{abs}, ΔQ, 1D S and DW

Band states: pending, for a hybrid later (F3 lists them)

A charge state passes? (ZPL ≥ excite_min, still on the defect) — no → Dropped unless native (why: summary.txt)

yes

(E)

D7: which excitations go on
- levels: bands localised on the defect (participation number at most local_fraction of the atoms), in the gap or up to excite_reach inside a band, where PBE's narrow gap pushes many
- the excitation: in each spin channel, the highest occupied level to the lowest empty one; CASTEP's ΔSCF from the ground state's .check, relaxed
- 1D estimates: $E_{rel} = E_{abs} - ZPL$, $\hbar\omega = \hbar v(2E_{rel})/\Delta Q$, $S = E_{rel}/\hbar\omega$, $DW = e^{-S}$ (a lower bound)
- a defect that passes goes on with all its charge states; native defects always do

Test D7 on one defect first
CASTEP's ΔSCF must work on your build: in CaO, Sb_{Ca}–V_O at q = −1 has a ZPL of about 0.53 eV (ADAQ). The thresholds are loose on purpose: the energies are PBE's.

E Production ground and excited states

E_production/ · the electronic cell, every charge state

(D)

E1 Production relaxations
every charge state of each defect D passes on, at A1's settings, in the electronic cell; the pristine cell too

Did a job fail? — yes → Flagged, left out of F: fix it and rerun with --resubmit-failed

no

E2 The ground states again
production settings, the bands' weights: the same excitation, found again

Its bands keep their numbers at every k-point? (spread < band_spread_tol) — no → Use a Γ-only ground state instead, at the production cut-off

yes

E2 Production ΔSCF
relaxed: ZPL, E_{abs}, ΔQ, 1D S and DW, with D7's verdicts; the excited structures go on to G

(F)

How the CASTEP jobs run
Every CASTEP step hands its jobs to supercell_convergence.py:
- one short dry run per distinct input sizes the jobs: cores or nodes, memory, partition and walltime; where nodes are shared and charged by the core (Cirrus), each job asks for just the cores it needs
- no more than max_jobs are queued at once
- a job that runs out of time carries on from its .check (up to max_restarts times; screening jobs don't restart)
- with tolerate_failures (the screening, hybrid and production steps), a failed job is flagged and the others carry on
- a finished job whose inputs have since changed stops the run, so results from two different settings never mix
- touch <workdir>/STOP stops dwf.py; queued jobs keep running

Stopping and rerunning
<workdir>/state.json records each finished step, with a fingerprint of its settings, input files and earlier results.
- stopped (STOP, Ctrl-C, --nos-wait, a crash)? Run the same command again: finished steps are skipped, and jobs already submitted or finished are picked up
- changed a setting? The steps that depend on it run again; jobs whose inputs are unchanged are reused
- --redo STEP: start that step again (its old files are kept)
- --status: what has finished; --until B: stop after phase B
- --prepare-only: write the job inputs, but submit nothing

Charge corrections: where they're applied
Lany–Zunger (default) or Makov–Payne: a point charge q screened by ε_0 (A2.3), in each cell's own shape (charge_correction).
- C3: the hull, stable charge states and binding (screening cell)
- D8: each size's $E_D - E_{host} + E_{corr}$
- F1–F2: the electronic cell

Not included: potential alignment, an anisotropic ε (averaged).

Why E2 checks the band numbers
CASTEP's ΔSCF constraints name a band number and spin channel, the same at every k-point. A level that disperses, or crosses a host band somewhere, would move the electron into different states at different k-points. In the electronic cell the band edges fold onto Γ and a localised level is flat, so the check passes; where it fails, E2 runs at Γ, where the bands are unambiguous.

Why PBE from E on
CASTEP's ΔSCF does not support hybrid functionals, so E2 and G stay at PBE (A1's settings). The hybrid is used where it decides the structure (D1–D5); the finalists from G go on to better methods (MBPT, or a hybrid ΔSCF elsewhere).

F Thermodynamics

F_thermo/ · every defect that reached E

(E)

F1 Charge corrections
Lany–Zunger or Makov–Payne (charge_correction), in the electronic cell

F2 Formation energies
at each chemical-potential limit, and the transition levels; plotted in formation.png

Find the Fermi level? (fermi_temperature) — yes → The self-consistent Fermi level of each dopant set (native, each dopant, each mix); quench_temperature: also after a quench

no

F3: an E2 excitation whose charge state forms in the gap? — no → End: formation energies, levels and Fermi levels

yes

(G)

G Optical lineshapes

G_lineshape/ · the excitations F3 sends on

(F)

G1 Configuration coordinate
ground-state and ΔSCF single points along the line between the relaxed states; parabolas: ħω and S of each, and the 1D lineshape

Multi-mode lineshape? (lineshape) — no → End: 1D lineshapes

yes

G2 A PBE model per defect
trained on displaced single points (production settings) and E1's steps; one head, the charge embedded

G2 Phonons in a big cell
finite differences with the model; ΔF (or ΔR) on each mode: the partial Huang–Rhys factors S_k

G2 Multi-mode lineshape
S(ħω) and the generating function, at lineshape_temperature

End: lineshapes

Phase G, and after it
PBE lineshapes in the equal-mode approximation (Alkauskas et al. 2014; Razinkovas et al. 2021). The finalists then go on to better methods, outside this workflow.